

Daniel Oppenheimer

oppenheimerd1@gmail.com | github.com/DannyOppenheimer | linkedin.com/in/daniel-oppenheimer | dannyooppenheimer.com

Education

University of Virginia

Aug 2023 – May 2027

- *BS in Computer Science, Data Science & Engineering Business Minor*
- Cumulative GPA: 3.78
- Relevant Coursework: Cloud Computing, Parallel Computing, Machine Learning, Artificial Intelligence, Computer Networks, Natural Language Processing, New Product Development, Data Science Systems
- Activities and Societies: Trigon Engineering Society @ UVA, Sigma Alpha Mu – Special Events Chair, 2025 Deloitte National Undergraduate Case Study Competition, Intramural Sports Supervisor

Experience

Comcast, Software Engineer Intern

June 2026 – August 2026

- Streamlined Xfinity's CI/CD migration to GitHub Actions and fixed flaky tests, reducing development build failures by 20% and eliminating hour-long release delays for a customer-facing product serving 30M users.
- Improved technical infrastructure for a team of ~50 by reducing API log clutter with reusable batching system and building tools to update stale development routes/allow for production payment flow testing.
- Architected prototype Wi-Fi troubleshooting menu for Xfinity app to drastically reduce frivolous customer support contacts and customer churn rate – earned commendation from Comcast CTO.

Booz Allen Hamilton, AI Software Engineer Intern

June 2025 – August 2025

- Engineered and benchmarked three distinct Retrieval-Augmented Generation (RAG) frameworks for a NOAA search tool, collaborating with their developers to enhance retrieval from a complex data storage system.
- Implemented a novel hybrid search system integrating BM25 keyword ranking with semantic search, leveraging a Reciprocal Rank Fusion pipeline and a genetic algorithm to optimize search result relevancy.
- Developed a full-stack solution with a Flask frontend, connecting to a backend on AWS EC2. Used REST API strategies to connect to NOAA data, a Redis cache, and the Gemma3 GenAI model.
- Developed CoT system to auto-upload new data formats in a metadata-rich format.

Zbooni, Software Development Consultant

Dec 2025 – Jan 2026

- Engineered a centralized fraud detection dashboard to automate risk assessment for the compliance team, replacing manual oversight with robust security rules, saving KYC team members 3 hours per workday.
- Architected an automated email verification workflow using Trengo for suspicious transactions, streamlining customer confirmation responses and significantly reducing the platform's exposure to fraudulent activity.
- Prototyped Random Forest and XGBoost models for risk data collected through platform in 6 months.

Glass House Recycling LLC, Co-founder

Feb 2020 – August 2025

- Founded curbside glass-pickup business at age 14 to meet demand after Falls Church City stopped providing municipal glass recycling service.
- Developed in-house invoicing system with a Node.js express web application.
- Leveraged digital advertising and print/radio media (NBC, WAMU, NowThis) to grow customer base to over 160 residential households, online stories about business reached 550k+ views, with 5k+ reactions.
- \$10k+ exit, selling company and converting over 80% of customers to the new platform.

Projects

Bracket Hub

Apr 2024 – Nov 2025

- Built social web application to bring a March Madness pointing system to user-made tournament brackets.
- Used React.js interfaced with Firebase authentication and Firestore database for backend and data storage.

Citi-Defense

May 2026 – Present

- Engineered 1-bit tower-defense game for an 8 MHz Macintosh Plus with only 1 MB of RAM.
- Built memory-efficient sprite compositor and asset pipeline using fixed-point, integer-only systems.

Bide

June 2026 – Present

- Built full-stack iOS application to surface better ETA and carpool information for friends/family meetups, solving the age-old “leaving 20 minutes early while your friend’s flight is 30 minutes delay” issue.
- Integrated PostgreSQL, Apple MapKit, In-App purchases, and iMessage tiles for an ideal user experience.

Spyfall

Jan 2020 – Apr 2020

- Designed minimalist “Spyfall” game during COVID lockdowns using Node.js + Socket.io for live multiplayer.
- Received GitHub “star” and fork on its public repository by a Senior Developer @ Microsoft.

Skills

Programming Languages:

Python, TypeScript/JavaScript, SQL, Java, HTML/CSS, C

Tools & Technologies:

React.js, Node.js, git, PyTorch, Unix, Cuda, GitHub Actions CI/CD, Ollama, Firebase, Flask, Django, Agile Method, Blender

Awards:

Eagle Scout